

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

The de Broglie wavelength (λ) of a particle is given by the formula :

$$\lambda = h / p$$

where h is the Planck's constant and p is the momentum of the particle .

The kinetic energy (E) of the electron is given by :

$$E = (1/2) m v ^ 2$$

where m is the mass of the electron and v is its velocity .

We can find the momentum (p) using the following equation :

$$p = m v$$

Substituting the values of m and v , we get :

$$p = \sqrt{ (2 m E) }$$

Now , we can plug in the values :

$$h = 6 . 626 \times 10 ^{-34} \text{ J s (Planck's constant)}$$

$$m = 9 . 109 \times 10 ^{-31} \text{ kg (mass of an electron)}$$

$$E = 1 \text{ eV} = 1 . 602 \times 10 ^{-19} \text{ J}$$

$$p = \sqrt{ (2 \times 9 . 109 \times 10 ^{-31} \text{ kg} \times 1 . 602 \times 10 ^{-19} \text{ J}) }$$

$$p = \sqrt{ (2 . 9218 \times 10 ^{-49} \text{ kg} ^ 2 \text{ m} ^ 2 / \text{s} ^ 2) }$$

$$p = 1 . 707 \times 10 ^{-24} \text{ kg m/s}$$

Now , we can find the de Broglie wavelength (λ):

$$\lambda = h / p$$

$$\lambda = (6 . 626 \times 10 ^{-34} \text{ J s}) / (1 . 707 \times 10 ^{-24} \text{ kg m/s})$$

$$\lambda = 3 . 88 \times 10 ^{-10} \text{ m}$$

Converting the wavelength to centimeters :

$$\lambda = 3 . 88 \times 10 ^{-10} \text{ m} \times (100 \text{ cm} / 1 \text{ m})$$

$$\lambda = 3 . 88 \times 10 ^{-8} \text{ cm}$$

Since there's no answer option match for $\lambda = 3 . 88 \times 10 ^{-8}$ cm we have reduced the choice list and

Go close as it had one variable

It quite far âGjassistant